Complex numbers and trigonometry

Lecture and worked examples

Objectives

Establish the addition and double-angle formulas; write a complex number in exponential form and choose the form suited to a computation; compute powers; use de Moivre's and Euler's formulas to transform trigonometric expressions.

Introductory activity

For every real number θ, let $f(\theta) = \cos\theta + i\sin\theta$.

Introductory activity (2)

1. Compute $f(0)$, $f\left(\frac{\pi}{2}\right)$, $f(\pi)$. What is the modulus of $f(\theta)$? Where does the image of $f(\theta)$ lie?
2. Write $f\left(\frac{\pi}{3}\right)$ and $f\left(\frac{\pi}{6}\right)$ in algebraic form, then compute the product $f\left(\frac{\pi}{3}\right) f\left(\frac{\pi}{6}\right)$. Compare it with $f\left(\frac{\pi}{2}\right)$.
3. Compute likewise $f\left(\frac{\pi}{4}\right)^2$ and compare it with $f\left(\frac{\pi}{2}\right)$.

Introductory activity (3)

4. With a calculator or a spreadsheet, compare $f(a)f(b)$ and $f(a+b)$ for three pairs (a, b) chosen at random.
5. Which known function satisfies a relation of the same kind, $g(a)g(b) = g(a+b)$ and $g(0) = 1$?
6. Expand $f(a)f(b)$. Which trigonometric formulas would one need to know to prove the conjecture?

Addition and double-angle formulas

For all real numbers a and b:

$$\cos(a-b) = \cos a \cos b + \sin a \sin b, \qquad \cos(a+b) = \cos a \cos b - \sin a \sin b,$$

$$\sin(a-b) = \sin a \cos b - \cos a \sin b, \qquad \sin(a+b) = \sin a \cos b + \cos a \sin b.$$

Addition and double-angle formulas

In a direct orthonormal frame, let $\overrightarrow{p}\begin{pmatrix}\cos a\\ \sin a\end{pmatrix}$ and $\overrightarrow{q}\begin{pmatrix}\cos b\\ \sin b\end{pmatrix}$: these vectors are unit vectors and $\left(\overrightarrow{u},\overrightarrow{p}\right)\equiv a$, $\left(\overrightarrow{u},\overrightarrow{q}\right)\equiv b \pmod{2\pi}$. By Chasles' relation, $\left(\overrightarrow{q},\overrightarrow{p}\right)\equiv a-b \pmod{2\pi}$. We compute $\overrightarrow{q}\cdot\overrightarrow{p}$ in two ways:

$$\overrightarrow{q}\cdot\overrightarrow{p}=\left\|\overrightarrow{q}\right\|\left\|\overrightarrow{p}\right\|\cos(a-b)=\cos(a-b)\quad\text{and}\quad\overrightarrow{q}\cdot\overrightarrow{p}=\cos b\cos a+\sin b\sin a.$$

This proves the first formula. Replacing b by $-b$ and using $\cos(-b)=\cos b$, $\sin(-b)=-\sin b$, we obtain $\cos(a+b)$. Finally $\sin(a+b)=\cos\left(\frac{\pi}{2}-a-b\right)=\cos\left(\left(\frac{\pi}{2}-a\right)-b\right)$, which the first formula expands into $\sin a\cos b+\cos a\sin b$; we deduce $\sin(a-b)$ by replacing b by $-b$.

Addition and double-angle formulas

For every real number a: $\sin 2a = 2 \sin a \cos a$ and
$\cos 2a = \cos^2 a - \sin^2 a = 2 \cos^2 a - 1 = 1 - 2 \sin^2 a$.

Addition and double-angle formulas

$$\cos\frac{\pi}{12} = \cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \frac{1}{2} \times \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} = \frac{\sqrt{6}+\sqrt{2}}{4}.$$

Product and arguments

For all real numbers θ and θ': $(\cos\theta + i\sin\theta)(\cos\theta' + i\sin\theta') = \cos(\theta+\theta') + i\sin(\theta+\theta')$.

Consequently, for all nonzero complex numbers z, z' and every integer n:

$$\arg(zz') \equiv \arg z + \arg z', \qquad \arg\frac{1}{z} \equiv -\arg z,$$

$$\arg\frac{z}{z'} \equiv \arg z - \arg z', \qquad \arg(z^n) \equiv n\arg z \pmod{2\pi}.$$

Product and arguments

Expanding gives $(\cos\theta\cos\theta' - \sin\theta\sin\theta') + i(\sin\theta\cos\theta' + \cos\theta\sin\theta')$, which the addition formulas simplify. If $z = r(\cos\theta + i\sin\theta)$ and $z' = r'(\cos\theta' + i\sin\theta')$, then $zz' = rr'(\cos(\theta+\theta') + i\sin(\theta+\theta'))$ with $rr' > 0$: this is a trigonometric form of zz'. The other relations follow: $z \times \dfrac{1}{z} = 1$ has argument 0, then $\dfrac{z}{z'} = z \times \dfrac{1}{z'}$, and an induction for $n \geq 0$, extended to $n < 0$ through the inverse.

Product and arguments

Multiplying by $\cos\theta + i\sin\theta$ turns through an angle θ about O. The conjecture of the activity of chapter 2 is proved: multiplying by i is a direct quarter turn, multiplying by $1 + i = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$ is an eighth of a turn followed by an enlargement with ratio $\sqrt{2}$.

Exponential form

For every real number θ, we write $\mathrm{e}^{i\theta} = \cos\theta + i\sin\theta$. Every nonzero complex number can be written $z = r\mathrm{e}^{i\theta}$ with $r = |z|$ and $\theta \equiv \arg z \pmod{2\pi}$: this is its *exponential form*.

Exponential form

For all real numbers θ, θ' and every integer n:

$$e^{i\theta}e^{i\theta'} = e^{i(\theta+\theta')}, \qquad \frac{1}{e^{i\theta}} = e^{-i\theta} = \overline{e^{i\theta}}, \qquad \frac{e^{i\theta}}{e^{i\theta'}} = e^{i(\theta-\theta')}, \qquad \left(e^{i\theta}\right)^n = e^{in\theta}.$$

Values to know: $e^{i0} = 1$, $e^{i\pi/2} = i$, $e^{i\pi} = -1$, $e^{-i\pi/2} = -i$, $\left|e^{i\theta}\right| = 1$.

Exponential form

For every real number θ and every integer n: $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$.

Exponential form

For every real number θ: $\cos\theta = \dfrac{e^{i\theta} + e^{-i\theta}}{2}$ and $\sin\theta = \dfrac{e^{i\theta} - e^{-i\theta}}{2i}$.

Exponential form

$\mathrm{e}^{i\theta} + \mathrm{e}^{-i\theta} = \mathrm{e}^{i\theta} + \overline{\mathrm{e}^{i\theta}} = 2\,\mathrm{Re}\left(\mathrm{e}^{i\theta}\right) = 2\cos\theta$, and $\mathrm{e}^{i\theta} - \mathrm{e}^{-i\theta} = 2i\,\mathrm{Im}\left(\mathrm{e}^{i\theta}\right) = 2i\sin\theta$.

Exponential form

For a sum, the algebraic form; for a product, a quotient or a power, the exponential form.
Example: $1 + i\sqrt{3} = 2\mathrm{e}^{i\pi/3}$, so

$$\left(1 + i\sqrt{3}\right)^{10} = 2^{10}\mathrm{e}^{10i\pi/3} = 1{,}024\,\mathrm{e}^{-2i\pi/3} = -512 - 512\sqrt{3}\,i,$$

since $\dfrac{10\pi}{3} - 4\pi = -\dfrac{2\pi}{3}$.

Exponential form

Expand $(\cos\theta + i\sin\theta)^3$ with the binomial formula and identify the real part:
$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta = 4\cos^3\theta - 3\cos\theta$.

Exponential form

$$\cos^3\theta = \frac{\left(e^{i\theta} + e^{-i\theta}\right)^3}{8} = \frac{e^{3i\theta} + 3e^{i\theta} + 3e^{-i\theta} + e^{-3i\theta}}{8} = \frac{\cos 3\theta + 3\cos\theta}{4}.$$

Exponential form

To factorise a sum of exponentials, take out the exponential of the mean angle:

$$\mathrm{e}^{ia} + \mathrm{e}^{ib} = \mathrm{e}^{i\frac{a+b}{2}} \left(\mathrm{e}^{i\frac{a-b}{2}} + \mathrm{e}^{-i\frac{a-b}{2}} \right) = 2\cos\frac{a-b}{2}\,\mathrm{e}^{i\frac{a+b}{2}}.$$

Exponential form

$\mathrm{e}^{i\theta}$ is not a positive real number: its modulus is 1, but it can be negative ($\mathrm{e}^{i\pi} = -1$) or non-real. An expression $r\mathrm{e}^{i\theta}$ is an exponential form only if $r > 0$.